Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Weekly Progress Report — Week 8

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1 Introduction

Week 8 focused on resolving weave texture interference: **frequency-domain weave suppression via 2D Fourier optics**. In structured textiles—especially complex herringbone weaves and dense twills—the strong repeating pattern of interlocking threads acts as visual camouflage. In the spatial domain, normal yarn ribs generate sharp intensity variations that produce high morphological residual noise. This forces detection thresholds upward, causing faint yarn flaws (such as missing picks, thread voids, and subtle weft disruptions) to go undetected.

To overcome the limits of purely spatial filtering, we developed an automated frequency-domain filtering module (`src/preprocessing/fft_notch.py`) that selectively attenuates periodic background weave harmonics in the 2D Fourier domain before spatial morphology is applied.

2 2D Fourier Analysis of Periodic Weaves

A fabric image can be considered as the combination of a strictly periodic background yarn pattern, a gradual lighting envelope, and localized defect disruptions:

2.1 Separation of Weave and Flaw Energy

When transformed into the 2D frequency domain using the 2D Fast Fourier Transform (FFT), repeating thread structures concentrate their energy into sharp, isolated harmonic peaks corresponding to the thread density and direction.

In contrast, localized defects are physically compact, meaning their energy is spread broadly across many continuous frequency bands. By identifying and selectively rejecting the discrete harmonic peaks of the repeating weave, the background texture is suppressed while the broad-spectrum energy of the defect is preserved.

3 Automated Harmonic Peak Detection

Because thread spacing varies between fabric types and under mechanical loom tension, peak coordinates cannot be hardcoded. We engineered an unsupervised peak-finding algorithm:

- **Low-Frequency DC Blocking:** A circular guard ring is placed around the central zero-frequency component to prevent general illumination variations and shading from being mistaken for thread peaks.
- **Local Spectral Peak Search:** The log-magnitude spectrum is scanned to locate dominant local maxima that represent fundamental weave frequencies.
- **Conjugate Symmetry Pairing:** Because real-world images possess symmetric Fourier spectra, each detected peak is paired with its reflection across the origin to maintain balanced filtering and ensure reconstructed images remain purely real-valued.

4 Butterworth Cross-Notch Filter Design

Filter profile selection is critical for quality reconstruction:

- **Avoiding Ringing Artifacts:** Ideal notch filters have sharp rectangular cutoffs that cause severe ringing ripples (Gibbs phenomenon) in the spatial domain, creating false artificial edges around threads. Gaus-

sian notch filters, on the other hand, have very wide transition slopes that attenuate too much non-periodic defect signal.

- **Order-3 Butterworth Profile:** We synthesized an order-3 Butterworth cross-notch rejection filter. Its smooth, continuous frequency transition eliminates spatial ringing while maintaining a tight rejection band centered on the weave harmonics.
- **Cross-Notch Geometry:** The filter employs paired radial notches along both horizontal and vertical axes to simultaneously suppress warp-weft cross-interference and diagonal twill ribs.

5 Spatial Reconstruction and Morphological Handoff

The filtered Fourier spectrum is converted back to the spatial domain using the 2D Inverse FFT. The real component is extracted, normalized, and handed directly to the dual-scale morphological residual engine. By eliminating background weave rhythms before morphological opening and closing, localized yarn flaws stand out with sharp contrast against a flattened, quiet background.

6 Work Completed This Week

- Implemented the 2D Fourier frequency-domain model for periodic fabric textures.
- Built automated harmonic peak detection with central DC guard rings and conjugate symmetry pairing.
- Synthesized the order-3 Butterworth cross-notch rejection filter in `src/preprocessing/fft_notch.py`.
- Verified that 2D Inverse FFT reconstruction suppresses repetitive yarn patterns without introducing edge-ringing artifacts.
- Developed a frequency analysis script (`code/scripts/run_phase2_frequency_notch.py`) to visualize spectra, notch attenuation surfaces, and spatial reconstructions.

7 Plan for Next Week

With web alignment (Phase 1) and frequency-domain weave suppression (Phase 2) complete, the final phase will address:

- Developing multi-class physical defect taxonomy using Hu's invariant moments and inertia tensors.
- Constructing a deterministic decision tree to classify flaws into Warp, Weft, Hole, and Contaminant categories.
- Integrating alignment, frequency notch filtering, dual-scale morphology, and defect taxonomy into an end-to-end inspection pipeline.

End of Week 8 Report